JAY DHABUWALA

Senior AI Engineer | Agentic AI, LLM Systems & RAG | ML Infrastructure

Pune, India | +91 9106728889 | jay.professional29@gmail.com | linkedin.com/in/jay-dhabuwala-1597ab238 | github.com/djay29

PROFESSIONAL SUMMARY

Senior AI Engineer with 3+ years of experience (over two at Tata Elxsi) building enterprise agentic AI systems, advanced RAG pipelines, and large-scale ML infrastructure. Promoted for technical impact, now owning solution architecture, proposals, and effort/cost estimation for GenAI engagements alongside hands-on delivery. Specialized in multi-agent orchestration (LangChain/LangGraph), hybrid search, LLM evaluation and grounding, multilingual speech AI pipelines, and distributed deep-learning training on GPU clusters. Independently built and deployed a full-stack LLM call intelligence platform on AWS. Delivered 20% accuracy gains, 30% fewer hallucinations, and predictive systems over 190M+ samples.

CORE TECHNICAL SKILLS

LLM & Agentic AI: LangChain, LangGraph, Multi-agent Orchestration, Tool Calling, Hybrid RAG (BM25 + Vector Search), Structured Output Generation, Adaptive Query Routing, Tiered Model Routing, Prompt Engineering, Prompt Caching, LLM Evaluation, Hallucination Grounding, MCP; AWS Bedrock (Claude Sonnet, Claude Haiku, Amazon Nova), OpenAI API (GPT)
Speech & Audio AI: Speech-to-Text (Soniox, AWS Transcribe), Speaker Diarization, Multilingual Indic Speech (Hindi, Gujarati, Marathi, Telugu, Kannada), WER/CER Evaluation, FFmpeg Audio Preprocessing
ML & Deep Learning: PyTorch (Distributed Data Parallel), TCN-LSTM / CNN-LSTM time-series models, XGBoost / LightGBM / CatBoost, Captum (Integrated Gradients), EVT/GPD anomaly thresholds, MLflow
Computer Vision & Imaging: YOLOv8/v11, ByteTrack, OpenCLIP, Vision-Language Models (Claude, Moondream2), OpenCV, Color Science (CIELAB, CIEDE2000), Image Compositing, Generative Image Editing (FLUX.2)
Data & Distributed Compute: Databricks, PySpark, Dask, Zarr, 8xH100 GPU training, Unity Catalog, 240M+ row telemetry
Backend, Frontend & Automation: FastAPI, Flask-SocketIO, REST APIs, Webhooks, Pydantic, JWT Authentication, Role-Based Access Control (RBAC), Real-time Streaming, n8n (Production Mastery certified); Frontend: React, TypeScript, Tailwind CSS, Streamlit
Cloud & DevOps: AWS (Lambda, API Gateway, S3, DynamoDB, CloudFront, Secrets Manager, EC2, RDS, ElastiCache), AWS CDK (Infrastructure as Code), Docker, Railway, Vercel, CI/CD Pipelines
Databases & Programming: DynamoDB, MySQL, Redis, Pinecone, FAISS, Elasticsearch, ChromaDB; Python (Advanced), JavaScript, Node.js, SQL, Pandas, NumPy, pytest, Linux/Bash, Git

PROFESSIONAL EXPERIENCE

Senior AI Engineer, LLM & AI Systems | Tata Elxsi

Apr 2026 - Present

- Own end-to-end technical architecture for new GenAI and ML engagements, from design to delivery planning.
- Author technical proposals and lead effort estimation and costing for prospective AI engineering projects.
- Continue hands-on delivery of agentic LLM systems and ML infrastructure alongside architecture and proposal work.

Engineer, LLM & AI Systems | Tata Elxsi

Mar 2024 - Apr 2026

- Architected and deployed multi-agent LLM orchestration systems (LangChain/LangGraph) handling complex document and vehicle-diagnostics workflows.
- Designed hybrid RAG pipelines combining BM25 keyword search with vector similarity search, improving accuracy by 20% and reducing LLM hallucinations by 30%.
- Built production-grade tool-calling systems enabling LLMs to execute database queries, API calls, and file operations with error handling and retry logic.
- Developed a Flask + SocketIO real-time LLM backend serving 50+ concurrent users with streaming responses, message checkpointing, and conversation memory.
- Deployed scalable AI infrastructure on AWS (EC2, RDS, ElastiCache, S3) with automated scaling and health monitoring.

Python Developer | YOU Broadband India Ltd

Sept 2023 - Mar 2024

- Built scalable analytics and automation pipelines processing 100K+ customer records for BI and churn prediction.
- Developed Oracle-based ETL workflows (cx_Oracle) and CRM-integrated customer re-engagement systems improving retention metrics.
- Created executive dashboards analyzing monthly revenue and variance metrics for C-suite visibility.

KEY TECHNICAL PROJECTS

BMS Predictive Anomaly Detection, EV Battery Management (Global Automotive OEM)

PyTorch DDP | Databricks | 8xH100 GPUs | Time-Series Deep Learning | PySpark | MLflow

- Architected a production predictive anomaly-detection system over 190M+ sliding-window samples of multivariate battery telemetry (60-timestep input sequences).
- Trained TCN-LSTM / CNN-LSTM deep-learning models on 8xH100 GPUs via PyTorch Distributed Data Parallel, using Zarr-backed datasets with NVMe and shared-memory caching for high-throughput training.
- Engineered asymmetric anomaly thresholds using Extreme Value Theory (EVT/GPD) per temperature bin, plus distributed gradient-boosting models (XGBoost/LightGBM/CatBoost) via Dask on Databricks GPU clusters.
- Built a DTC confidence-scoring pipeline using Wilson lower-bound lift scoring across 240M+ rows in PySpark, a Captum Integrated Gradients explainability module, and MLflow experiment tracking with Unity Catalog I/O.

Call Insight Engine: AI Call Intelligence Platform (Independent Project)

Python | FastAPI | AWS Bedrock (Claude) | OpenAI GPT | Soniox STT | React | AWS Lambda | DynamoDB | AWS CDK

- Built and deployed an end-to-end platform that transcribes multilingual sales calls (Hindi, Gujarati, Marathi, Telugu, Kannada) and uses LLMs to summarize, extract CRM enquiry fields, and detect compliance red flags, pricing violations, and agent misconduct.
- Designed a tiered LLM pipeline (low-cost screening model, escalation to a stronger model on flags, low confidence, or long calls) with prompt caching and a grounding gate that rejects any flag whose verbatim evidence quote is not in the transcript.
- Benchmarked speech-to-text providers on real call audio and moved to Soniox, cutting STT cost ~14x versus AWS Transcribe (STT was ~85% of per-call cost); built a WER/CER evaluation harness and an 8 kHz telephony test corpus.
- Ran a pilot on 750+ real calls in one month at ~INR 0.15 per call; added call-connection detection that auto-filtered 90 of 181 non-conversation recordings (voicemail, busy tones, call screeners), all confirmed on manual review, and fixed an Indic-script validation bug that had inflated LLM spend by ~15%.
- Shipped JWT auth with 4-role RBAC, audit logging, a React dashboard (review queue, agent analytics, cost tracking), idempotent two-way CRM integration (REST + webhooks), and AWS infrastructure as code via CDK (Lambda, API Gateway, S3, DynamoDB, CloudFront); 300+ automated tests.

Drone Security System: Multimodal Agentic Video Surveillance (Independent Project)

Python | LangGraph | AWS Bedrock (Claude Haiku) | YOLOv8 + ByteTrack | OpenCLIP | ChromaDB | SQLite | Streamlit | Docker

- Built a real-time surveillance agent that ingests RTSP or recorded drone footage and runs a 6-node LangGraph pipeline (perceive, contextualize, rule check, LLM judge, alert, log) combining YOLOv8, ByteTrack tracking, CLIP embeddings, and VLM captions.
- Designed a hot-reloadable YAML rule engine (after-hours intrusion, loitering, gate tampering) with conditional LLM escalation: Claude Haiku (Bedrock) rules ambiguous alerts genuine or false positive; per-zone cooldowns deduplicate alerts.
- Kept detection real-time by running slower VLM captioning asynchronously in a thread pool with a local Moondream2 fallback for offline use, plus backpressure-aware streaming (drop oldest frames on live feeds, block on recorded files).
- Built hybrid retrieval over CLIP vectors (ChromaDB) and structured events (SQLite) with LLM intent parsing, powering a Streamlit dashboard with live alerts, an event timeline, and multi-turn natural-language search over footage; 120+ tests.

RangMatch: Embroidery Thread Colour-Matching Engine (Independent Project, Prototype)

JavaScript | Python | Color Science (CIELAB, CIEDE2000) | Optimization (Beam Search) | FLUX.2 (fal.ai)

- Built an offline, deterministic engine that parses embroidery design files (DST), reads fabric colour from a photo, and generates palettes snapped to a 183-shade stocked catalogue with exact thread numbers, never an unavailable shade.
- Developed a layout-aware palette fitter (beam search over adjacency and contrast constraints) that cut indistinguishable adjacent colour blocks from 16.2% to 0.7% across 148 designs x 7 fabrics x 4 styles, guided by a regional colour model built from 99 annotated images across 8 Indian states.
- Computed per-shade thread consumption (metres, cones) from stitch geometry (~3.6 m per 1,000 stitches, validated over 267K stitches) and rendered garment previews via procedural compositing plus an optional FLUX.2 finishing pass.

Automotive System & Software Requirements Automation (LLM-Based)

LangChain | Multi-agent Validation | ASPICE / Regulatory Compliance

- Engineered an automation pipeline generating complete ASPICE-compliant System and Software Requirements specifications using domain templates, regulatory standards, and constraint-based prompting.
- Reduced requirement-drafting time and improved consistency across engineering teams. Stack: LangChain, Claude.

Additional projects: Team Knowledge Enablement (S3-triggered n8n ingestion pipeline into Pinecone + RAG chatbot for cross-team product knowledge, FastAPI); News-to-Script LangGraph pipeline (virality scoring + short-form script generation); PPT automation (LangChain + MCP + python-pptx); LLM chatbot with tool calling, memory & checkpointing (FastAPI / MySQL / Redis).

CERTIFICATIONS

n8n Production Mastery | MCP Mastery (Claude, LangChain, Ollama) | Complete MLOps Bootcamp | FastAPI Advanced (Udemy)

EDUCATION

B.Tech, Computer Science | Visvesvaraya Technological University (VTU) | 2019 - 2023 | First Class Distinction (8 CGPA)

ACHIEVEMENTS & RECOGNITION

- Extra Mile Award, Tata Elxsi, twice: 2026 (BMS Anomaly Detection) and 2025 (high-impact LLM solutions).
- Multiple production systems deployed with 99.9% uptime SLA across automotive and enterprise deployments.
- State-Level Swimming Award (Gujarat).

LANGUAGES

English (Fluent) | Hindi (Fluent) | Gujarati (Native)